RAG CHATBOT DOCUMENTATION

• OVERVIEW:

This project implements a Retrieval-Augmented Generation (RAG) chatbot capable of answering user questions from SBI Mutual Fund Scheme Information Documents (SIDs). Instead of relying solely on the Large Language Model (LLM), the chatbot first retrieves relevant information from uploaded PDF documents and then uses the LLM to generate accurate, context-aware responses. The system reduces hallucinations by ensuring that answers are grounded in the retrieved document context.

• OBJECTIVE:

The main objectives of the project are:

- Extract information from multiple mutual fund PDF documents.
- Convert document text into vector embeddings.
- Retrieve the most relevant information using semantic similarity search.
- Generate human-readable answers using an instruction-tuned Large Language Model.
- Provide accurate responses while minimizing hallucinations.

• DATASET:

The chatbot uses the following SBI Mutual Fund Scheme Information Documents:

- SBI Arbitrage Opportunities Fund

- SBI Corporate Bond Fund
- SBI Dividend Yield Fund
- SBI Fixed Maturity Plan (Series 47)

• TECHNOLOGY STACK:

Sentence Transformers (all-MiniLM-L6-v2) is used to generate semantic embeddings for the extracted text. These embeddings are stored and searched efficiently using FAISS (Facebook AI Similarity Search) for semantic retrieval. The chatbot uses Qwen2.5-1.5B-Instruct, a transformer-based Large Language Model from Hugging Face, to generate context-aware answers based on the retrieved information.

• IMPLEMENTATION STEPS:

1. PDF LOADING:

The uploaded SBI Mutual Fund PDFs are read using **PyMuPDF**.

For every page:

- Page number is stored.
- Source document name is stored.
- Extracted text is stored.

2. CHUNK CREATION:

Instead of splitting text by characters, each PDF page is stored as a separate chunk.

Each chunk contains:

- Source filename
- Page number
- Page text

This preserves the complete information on each page.

3. EMBEDDING GENERATION:

Each chunk is converted into a dense vector representation using:

sentence-transformers/all-MiniLM-L6-v2

This model generates 384-dimensional embeddings.

4. FAISS INDEX:

All embeddings are stored inside a FAISS vector database. FAISS enables fast semantic similarity search. The chatbot retrieves the most relevant chunks based on cosine similarity.

5. RETREIVAL:

When a user asks a question:

Example:

What is the exit load of SBI Arbitrage Opportunities Fund?

The system:

- Converts the question into an embedding.
- Searches FAISS.
- Retrieves the most relevant document pages.
- Builds a context using retrieved pages.

6. PROMPT:

The retrieved context and user question are combined into a prompt. Only retrieved information is provided to the LLM.

7. Answer Generation:

The prompt is passed to:

“Qwen2.5-1.5B-Instruct”

The model generates a concise natural language answer using only the retrieved context.

- ## EXAMPLE QUESTIONS:
  - What is dividend fund?

A dividend fund, as described in the given text, is designed to capitalize on high-dividend-paying stocks. The scheme aims to invest primarily in companies with a high dividend yield, either through direct investments or through buybacks. Additionally, it considers companies that have paid dividends or engaged in buybacks during at least one of the previous three financial years. The goal is to create a portfolio with an average dividend yield significantly higher than that of the Nifty 50 index. The fund employs a bottom-up approach to stock selection, focusing on diverse portfolios across different market caps, sectors, and industries.

REFERENCES
SBI_Arbitrage_Opportunities_Fund.pdf | Page 12
SBI_Dividend_Yield_Fund.pdf | Page 12

- What are the major risks associated with SBI Corporate Bond Fund?

  The major risks associated with SBI Corporate Bond Fund include:

  1. **Credit Risk**: This involves the risk that the issuer might fail to make payments on time.

  2. **Liquidity Risk**: This refers to the difficulty in selling securities quickly when needed.

  3. **Interest Rate Risk**: Fluctuations in interest rates affect the value of bonds held by the fund.

  4. **Reinvestment Risk**: Uncertainty about the return on reinvestment of cash flows from bonds.

  5. **Tax Risks**: Potential changes in tax laws could affect the fund's performance.

  These risks are

  REFERENCES

SBI_Corporate_Bond_Fund.pdf | Page 12
SBI_Corporate_Bond_Fund.pdf | Page 22

- What securities can the fund invest in?

The fund can invest in various types of securities including but not limited to:

1. Floating rate debt instruments issued by central government, corporates, PSUs etc., with periodic coupon resets.
2. Repo (Repurchase Agreement) or Reverse Repo.
3. Securitized Debt (SD)/Pass Through Certificate (PTC).
4. Debt Derivatives.
5. Credit Default Swaps.
6. Repo transactions in corporate debt securities.

Additionally, the fund can invest in units of mutual funds, units issued by REITs/InvITs, foreign securities including ADR/GDR/

REFERENCES
SBI_Arbitrage_Opportunities_Fund.pdf | Page 12
SBI_Corporate_Bond_Fund.pdf | Page 12

- Who manages the fund?

Mr. Neeraj Kumar manages the SBI Arbitrage Opportunities Fund. He has been managing the scheme since October 2012 and has over 25 years of experience in equity dealing, equity research, and Finance & Accounts Department.

REFERENCES
SBI_Arbitrage_Opportunities_Fund.pdf | Page 13
SBI_Corporate_Bond_Fund.pdf | Page 13

- What benchmark is used by the scheme?

The first tier benchmark of the scheme is the Nifty Corporate Bond Index A-II.

REFERENCES
SBI_Arbitrage_Opportunities_Fund.pdf | Page 13
SBI_Corporate_Bond_Fund.pdf | Page 12

- What is the minimum investment amount?

The minimum investment amount for the Scheme is Rs. 5000/- and in multiples of Re. 1 thereafter.

- What are the taxation rules for SBI Dividend Yield Fund?

The document does not provide specific tax information for the SBI Dividend Yield Fund. It only mentions that the fund aims to generate opportunities for growth along with consistent returns through investments in dividend-yielding companies. For detailed tax rules, investors should consult with a tax advisor or refer to official tax guidelines.

REFERENCES

SBI_Dividend_Yield_Fund.pdf | Page 14

SBI_Dividend_Yield_Fund.pdf | Page 1

- Which schemes are open-ended and which are close-ended?

  Based on the given text, "Open-ended" refers to schemes where investors have the flexibility to redeem their holdings at any time without penalty. Examples include arbitrage opportunities funds, flex systematic transfer plans, and daily/weekly idcw options.

  "Semi-Open-ended" schemes are those where investors can redeem their holdings after a specific period has elapsed, typically three years. This contrasts with closed-end schemes, which do not offer such flexibility.

  Closed-end schemes are defined as those where investors cannot redeem their holdings until they reach maturity, usually within five years. Examples include mutual fund schemes like arbitrage opportunities

  REFERENCES

  SBI_Arbitrage_Opportunities_Fund.pdf | Page 54

  SBI_Arbitrage_Opportunities_Fund.pdf | Page 56

- Is the scheme listed on a stock exchange?

  Yes, the scheme is listed on a stock exchange.

- What is the investment objective of SBI Fixed Maturity Plan Series 47?

  The investment objective of the SBI Fixed Maturity Plan Series 47 is to provide regular income and capital growth with limited interest rate risk to investors through investments in a portfolio comprising of debt instruments such as Government Securities, PSU & Corporate Bonds, and Money Market Instruments maturing on or before the maturity of the scheme. There is no assurance or guarantee that the scheme's objective will be achieved.

  REFERENCES
  SBI_Fixed_Maturity_Plan_(FMP)-_Series_47_(1434_Days).pdf | Page 4
  SBI_Fixed_Maturity_Plan_(FMP)-_Series_47_(1434_Days).pdf | Page 5

- CONCLUSION:

This project demonstrates the implementation of a Retrieval-Augmented Generation (RAG) chatbot for answering questions from SBI Mutual Fund Scheme Information Documents. The system combines semantic retrieval using Sentence Transformers and FAISS with the Qwen2.5-1.5B-Instruct Large Language Model to generate context-aware responses. By retrieving relevant document sections before generation, the chatbot provides more reliable and document-grounded answers while reducing hallucinations. The modular design also allows additional documents and future enhancements to be incorporated easily.